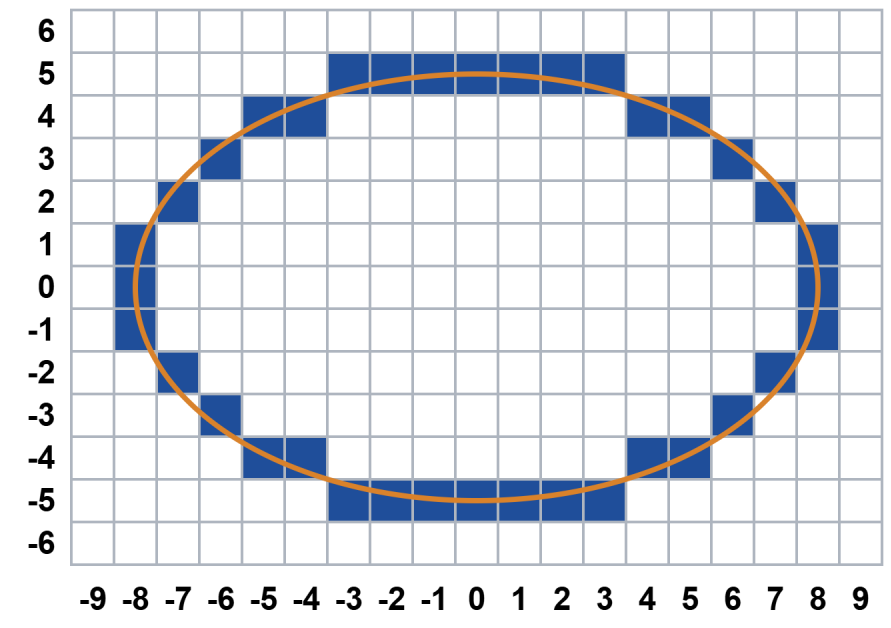
0613-403 · Week 3 · Lecture 6

The Midpoint Ellipse Algorithm

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Review problems: solve before we start

Q1 · DDA

Find the pixels of the line from (1, 2) to (7, 6).

Q2 · Bresenham

Find the pixels of the line from (3, 2) to (12, 7).

Q3 · Midpoint line

Find the pixels of the line from (0, 0) to (8, 3).

Q4 · Midpoint circle

Find the first-octant pixels of a circle with r = 9, centre (0, 0).

Show every step in a table, the way you will in the exam. Time: 12 minutes.

Q1: DDA, (1, 2) to (7, 6)

Step along the longer side; add xInc and yInc each step and round.

Step 1: dx = ____, dy = ____

Step 2: steps = max(|dx|, |dy|) = ____

Step 3: xInc = dx ÷ steps = ____

Step 4: yInc = dy ÷ steps = ____

Step 5: start at (1, 2), add the increments each step

Round with floor(v + 0.5).

k	x	y	Pixel (round x, round y)

Q1 solution: DDA, (1, 2) to (7, 6)

Step along the longer side; add xInc and yInc each step and round.

Step 1: dx = 7 − 1 = 6, dy = 6 − 2 = 4

Step 2: steps = max(|6|, |4|) = 6

Step 3: xInc = 6 ÷ 6 = 1

Step 4: yInc = 4 ÷ 6 = 0.67

Step 5: start at (1, 2), add the increments 6 times

Round with floor(v + 0.5).

k	x	y	Pixel (round x, round y)
0	1	2	(1, 2)
1	2	2.67	(2, 3)
2	3	3.33	(3, 3)
3	4	4	(4, 4)
4	5	4.67	(5, 5)
5	6	5.33	(6, 5)
6	7	6	(7, 6)

Q2: Bresenham, (3, 2) to (12, 7)

Integers only: one test and one addition per pixel.

Step 1: dx = ____, dy = ____

Step 2: d_1 = 2dy − dx = ____

Step 3: inc1 = 2dy = ____ (d < 0, choose S)

Step 4: inc2 = 2(dy − dx) = ____ (d ≥ 0, choose T)

Step 5: plot (x, y), test d, update d, x + 1

Check: dx + 1 pixels, last is (12, 7).

x	y	d	Decision	New d

Q2 solution: Bresenham, (3, 2) to (12, 7)

Integers only: one test and one addition per pixel.

Step 1: dx = 12 − 3 = 9, dy = 7 − 2 = 5

Step 2: d_1 = 2dy − dx = 10 − 9 = 1

Step 3: inc1 = 2dy = 10 (d < 0, choose S)

Step 4: inc2 = 2(dy − dx) = −8 (d ≥ 0, choose T)

Step 5: plot (x, y), test d, update d, x + 1

Check: dx + 1 = 10 pixels, last is (12, 7).

x	y	d	Decision	New d
3	2	1	d ≥ 0 → T, y + 1	-7
4	3	-7	d < 0 → S, y stays	3
5	3	3	d ≥ 0 → T, y + 1	-5
6	4	-5	d < 0 → S, y stays	5
7	4	5	d ≥ 0 → T, y + 1	-3
8	5	-3	d < 0 → S, y stays	7
9	5	7	d ≥ 0 → T, y + 1	-1
10	6	-1	d < 0 → S, y stays	9
11	6	9	d ≥ 0 → T, y + 1	1
12	7	—	Stop: reached (12, 7)	—

Q3: midpoint line, (0, 0) to (8, 3)

Test the midpoint M; d is F(M) scaled by 2.

Step 1: dx = ____, dy = ____

Step 2: $d_1 = 2dy - dx$ = ____

Step 3: ΔE = 2dy = ____ (d ≤ 0, choose E)

Step 4: ΔNE = 2(dy − dx) = ____ (d > 0, choose NE)

Step 5: plot, test d, update d, x + 1

Tie rule: at d = 0 choose E.

x	y	d	Decision	New d

Q3 solution: midpoint line, (0, 0) to (8, 3)

Test the midpoint M; d is F(M) scaled by 2.

Step 1: dx = 8, dy = 3

Step 2: $d_1 = 2dy - dx = 6 - 8 = -2$

Step 3: ΔE = 2dy = 6 (d ≤ 0, choose E)

Step 4: ΔNE = 2(dy − dx) = −10 (d > 0, choose NE)

Step 5: plot, test d, update d, x + 1

Note: at d = 0 midpoint picks E; Bresenham would pick T.

x	y	d	Decision	New d
0	0	-2	d ≤ 0 → E, y stays	4
1	0	4	d > 0 → NE, y + 1	-6
2	1	-6	d ≤ 0 → E, y stays	0
3	1	0	d ≤ 0 → E, y stays	6
4	1	6	d > 0 → NE, y + 1	-4
5	2	-4	d ≤ 0 → E, y stays	2
6	2	2	d > 0 → NE, y + 1	-8
7	3	-8	d ≤ 0 → E, y stays	-2
8	3	—	Stop: reached (8, 3)	—

Q4: midpoint circle, r = 9

Compute one octant (x from 0 up to y), then use 8-way symmetry.

Step 1: start at (0, r) = ____

Step 2: $p_0 = 1 - r$ = ____

Step 3: $p < 0 \rightarrow$ E: $p + 2x + 3$

Step 4: $p \geq 0 \rightarrow$ SE: $p + 2(x - y) + 5$, $y - 1$

Step 5: $x + 1$ each step; stop when $x > y$

(x, y are the values before the step.)

x	y	p	Decision	New p

Q4 solution: midpoint circle, r = 9

Compute one octant (x from 0 up to y), then use 8-way symmetry.

Step 1: start at (0, r) = (0, 9)

Step 2: $p_0 = 1 - r = 1 - 9 = -8$

Step 3: p < 0 → E: p + 2x + 3

Step 4: p ≥ 0 → SE: p + 2(x − y) + 5, y − 1

Step 5: x + 1 each step; stop when x > y

(x, y are the values before the step.) Next would be (7, 6): x > y, so stop.

x	y	p	Decision	New p
0	9	-8	p < 0 → E, y stays	-5
1	9	-5	p < 0 → E, y stays	0
2	9	0	p ≥ 0 → SE, y − 1	-9
3	8	-9	p < 0 → E, y stays	0
4	8	0	p ≥ 0 → SE, y − 1	-3
5	7	-3	p < 0 → E, y stays	10
6	7	10	p ≥ 0 → SE, y − 1	13

Warm-up: midpoint circle in 3 minutes

Q1

Circle with r = 5. What is p_0?

Q2

Write the 8 symmetric points of (2, 5).

Q3

Why do we stop when x > y?

Today: the same midpoint idea for an ellipse, a circle stretched in one direction.

Today's plan

First

Review problems

- DDA
- Bresenham
- Midpoint line
- Midpoint circle
- Step-by-step tables

Part 1

Midpoint ellipse

- 4-way symmetry
- Two regions
- Proof of every formula
- Worked example and exercise
- C++ code

Part 1

The midpoint ellipse algorithm

A circle stretched in one direction

The ellipse equation

Centre at the origin

$$x^2/a^2 + y^2/b^2 = 1$$

a = half-width (along x), b = half-height (along y).

a = b gives a circle.

Without fractions

Multiply both sides by $a^2 \cdot b^2$:

$$b^2x^2 + a^2y^2 = a^2b^2$$

Centre (xc, yc): draw at (0, 0), then add xc and yc to every pixel.

The ellipse function

$$F(x, y) = b^2x^2 + a^2y^2 - a^2b^2$$

$F < 0$

Point is inside the ellipse

$F = 0$

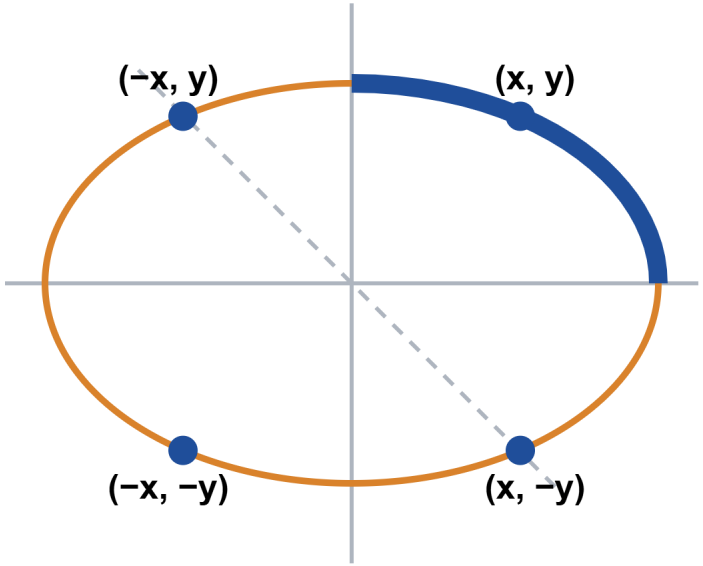
Point is on the ellipse

$F > 0$

Point is outside the ellipse

Same test as the circle: put the midpoint into F and look at the sign.

4-way symmetry (not 8)



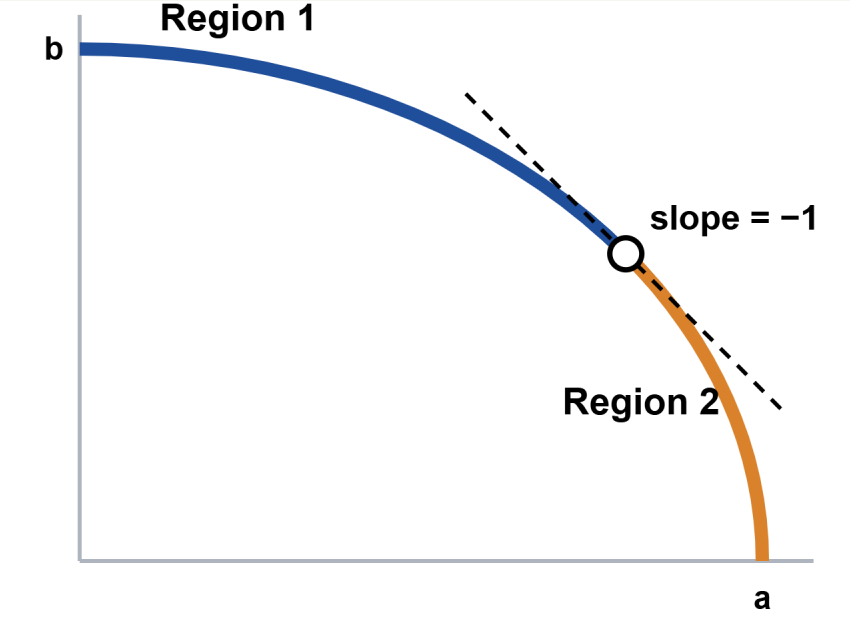
An ellipse is symmetric about the x-axis and the y-axis only.

(±x, ±y): one point gives 4 pixels

It is not symmetric about y = x (the dashed line), because a ≠ b.
Swapping x and y would give a different ellipse.

So we compute a whole quadrant (blue): from (0, b) to (a, 0).

Two regions in the quadrant



Region 1 (blue): |slope| < 1

Flat part. Step x by 1 every time; y stays or goes down 1.

Choices: E or SE.

Region 2 (orange): |slope| > 1

Steep part. Step y down by 1 every time; x stays or goes up 1.

Choices: S or SE.

Why two? The same reason as steep lines: always step along the longer direction, or pixels leave gaps.

Every variable and its meaning

Symbol	Meaning
a, b	Radius along x and along y (integers)
(xc, yc)	Centre of the ellipse
(x_k, y_k)	Current pixel, measured from the centre
F(x, y)	Ellipse function: $b^2x^2 + a^2y^2 - a^2b^2$
$p1_k$	Region 1 decision: F at the midpoint $(x_k + 1, y_k - ½)$
$p2_k$	Region 2 decision: F at the midpoint $(x_k + ½, y_k - 1)$
$2b^2x$, $2a^2y$	Running values: compare them to know when region 1 ends

Proof 1: where region 1 ends

Differentiate $b^2x^2 + a^2y^2 = a^2b^2$ with respect to x:

$2b^2x + 2a^2y \cdot (dy/dx) = 0$

$dy/dx = -2b^2x \div 2a^2y$

$|\text{slope}| < 1$ while $2b^2x < 2a^2y$

Stay in region 1 while $2b^2x < 2a^2y$

Why this is cheap

$2b^2x$ goes up by $2b^2$ each time x goes up.

$2a^2y$ goes down by $2a^2$ each time y goes down.

So the test needs only additions.

Proof 2: the region 1 decision

Choices: $E = (x_k + 1, y_k)$ and $SE = (x_k + 1, y_k - 1)$

Midpoint between them: $(x_k + 1, y_k - ½)$

$p1_k = F(x_k + 1, y_k - ½)$

$= b^2(x_k + 1)^2 + a^2(y_k - ½)^2 - a^2b^2$ ... (1)

$p1_k < 0 \rightarrow E$; $p1_k \geq 0 \rightarrow SE$

Meaning

$p1_k < 0$: the midpoint is inside, so the ellipse passes above it. E is closer.

$p1_k \geq 0$: the midpoint is outside or on it. SE is closer.

Proof 3: region 1 update

Next: $p1_{k+1} = b^2(x_{k+1} + 1)^2 + a^2(y_{k+1} - \tfrac{1}{2})^2 - a^2b^2$, with $x_{k+1} = x_k + 1$

Subtract (1): $p1_{k+1} = p1_k + 2b^2x_{k+1} + b^2 + a^2[(y_{k+1} - \tfrac{1}{2})^2 - (y_k - \tfrac{1}{2})^2]$

E ($y_{k+1} = y_k$): the bracket is 0

SE ($y_{k+1} = y_k - 1$): the bracket is $-2y_{k+1}$

E: $p1_{k+1} = p1_k + 2b^2x_{k+1} + b^2$

SE: $p1_{k+1} = p1_k + 2b^2x_{k+1} - 2a^2y_{k+1} + b^2$

Check the x-part

$b^2[(x_k + 2)^2 - (x_k + 1)^2]$

$= b^2(2x_k + 3)$

$= 2b^2(x_k + 1) + b^2 = 2b^2x_{k+1} + b^2$

Proof 4: the starting value $p1_0$

First pixel: $(0, b)$. First midpoint: $(1, b - \tfrac{1}{2})$

$p1_0 = b^2(1)^2 + a^2(b - \tfrac{1}{2})^2 - a^2b^2$

$= b^2 + a^2(b^2 - b + \tfrac{1}{4}) - a^2b^2$

$= b^2 + a^2b^2 - a^2b + \tfrac{1}{4}a^2 - a^2b^2$

$$p1_0 = b^2 - a^2b + \tfrac{1}{4}a^2$$

Example

$a = 8, b = 6$:

$p1_0 = 36 - 64(6) + \tfrac{1}{4}(64)$

$= 36 - 384 + 16 = -332$

Proof 5: region 2 decision and update

Choices: $S = (x_k, y_k - 1)$ and $SE = (x_k + 1, y_k - 1)$

Midpoint: $(x_k + \frac{1}{2}, y_k - 1)$; $p2_k = F(x_k + \frac{1}{2}, y_k - 1)$

$p2_k > 0 \rightarrow$ midpoint outside $\rightarrow S$; $p2_k \leq 0 \rightarrow SE$

Same subtraction as before, now y always goes down by 1:

S: $p2_{k+1} = p2_k - 2a^2y_{k+1} + a^2$

SE: $p2_{k+1} = p2_k + 2b^2x_{k+1} - 2a^2y_{k+1} + a^2$

Check the y-part

$a^2[(y_k - 2)^2 - (y_k - 1)^2]$

$= a^2(-2y_k + 3)$

$= -2a^2y_{k+1} + a^2$

x-part for SE: $b^2[(x_k + 3/2)^2 - (x_k + \frac{1}{2})^2] = 2b^2x_{k+1}$

Proof 6: the starting value $p2_0$

Region 2 starts at the last region 1 pixel (x_0, y_0).

Its first midpoint is $(x_0 + ½, y_0 − 1)$:

$$p2_0 = b^2(x_0 + ½)^2 + a^2(y_0 − 1)^2 − a^2b^2$$

Compute it once, directly from F, then keep updating.

Example

$a = 8$, $b = 6$, last region 1 pixel $(7, 3)$:

$p2_0 = 36(7.5)^2 + 64(2)^2 − 2304$

$= 2025 + 256 − 2304 = −23$

Formula card: midpoint ellipse

Region 1: start at (0, b)

$$p1_0 = b^2 - a^2 b + \tfrac{1}{4}a^2$$

p1 < 0 → E:

$$p1 + 2b^2 x_{k+1} + b^2$$

p1 ≥ 0 → SE:

$$p1 + 2b^2 x_{k+1} - 2a^2 y_{k+1} + b^2$$

Stay while $2b^2 x < 2a^2 y$

Region 2: start at last pixel (x_0, y_0)

$$p2_0 = b^2(x_0 + \tfrac{1}{2})^2 + a^2(y_0 - 1)^2 - a^2 b^2$$

p2 > 0 → S:

$$p2 - 2a^2 y_{k+1} + a^2$$

p2 ≤ 0 → SE:

$$p2 + 2b^2 x_{k+1} - 2a^2 y_{k+1} + a^2$$

Stop when y = 0

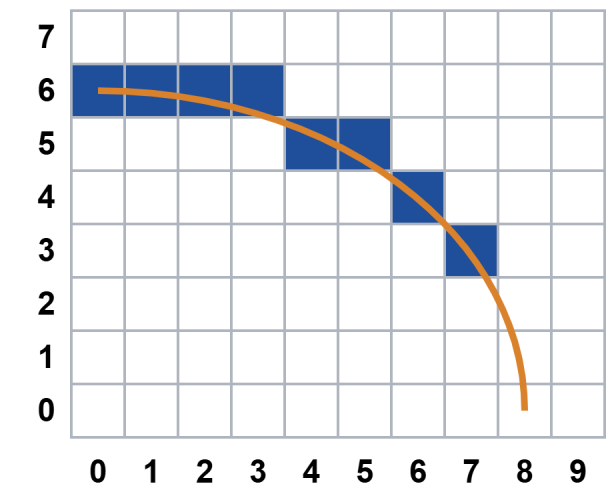
Midpoint ellipse algorithm

1. Input a, b and centre (xc, yc). Start at x = 0, y = b.
2. p1 = $b^2 - a^2b + \frac{1}{4}a^2$; px = 0; py = $2a^2y$. Plot the 4 symmetric points.
3. Region 1, while px < py: x = x + 1, px = px + $2b^2$. If p1 < 0: p1 = p1 + px + b^2. Else: y = y − 1, py = py − $2a^2$, p1 = p1 + px − py + b^2. Plot 4 points.
4. p2 = $b^2(x + \frac{1}{2})^2 + a^2(y - 1)^2 - a^2b^2$
5. Region 2, while y > 0: y = y − 1, py = py − $2a^2$. If p2 > 0: p2 = p2 − py + a^2. Else: x = x + 1, px = px + $2b^2$, p2 = p2 + px − py + a^2. Plot 4 points.
6. Every plotted point is shifted by (xc, yc).

Example: a = 8, b = 6 (region 1)

Start (0, 6). $p1_0 = -332$. $2b^2 = 72$, $2a^2 = 128$. Start: $2a^2y = 768$.

k	$p1_k$	Decision	(x_{k+1}, y_{k+1})	$2b^2x_{k+1}$	$2a^2y_{k+1}$
0	-332	p1 < 0 → E	(1, 6)	72	768
1	-224	p1 < 0 → E	(2, 6)	144	768
2	-44	p1 < 0 → E	(3, 6)	216	768
3	208	p1 ≥ 0 → SE	(4, 5)	288	640
4	-108	p1 < 0 → E	(5, 5)	360	640
5	288	p1 ≥ 0 → SE	(6, 4)	432	512
6	244	p1 ≥ 0 → SE	(7, 3)	504	384

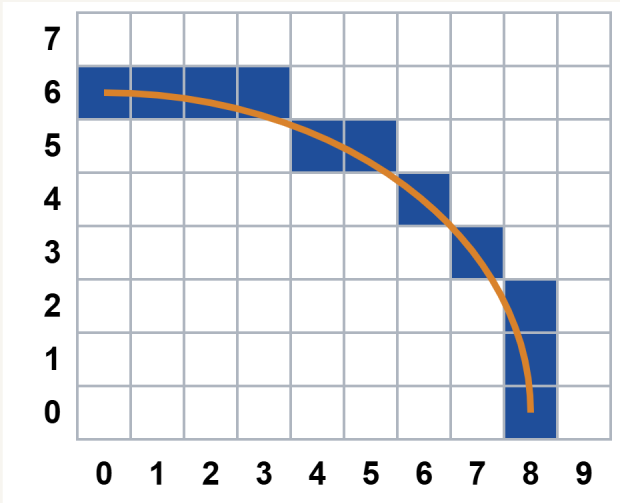


Example: a = 8, b = 6 (region 2)

Start at (7, 3). $p2_0 = 36(7.5)^2 + 64(2)^2 - 2304 = -23$.

k	$p2_k$	Decision	(x_{k+1}, y_{k+1})	$2b^2x_{k+1}$	$2a^2y_{k+1}$
0	-23	p2 ≤ 0 → SE	(8, 2)	576	256
1	361	p2 > 0 → S	(8, 1)	576	128
2	297	p2 > 0 → S	(8, 0)	576	0

Quadrant pixels: (0,6) (1,6) (2,6) (3,6) (4,5) (5,5) (6,4) (7,3) (8,2) (8,1) (8,0)



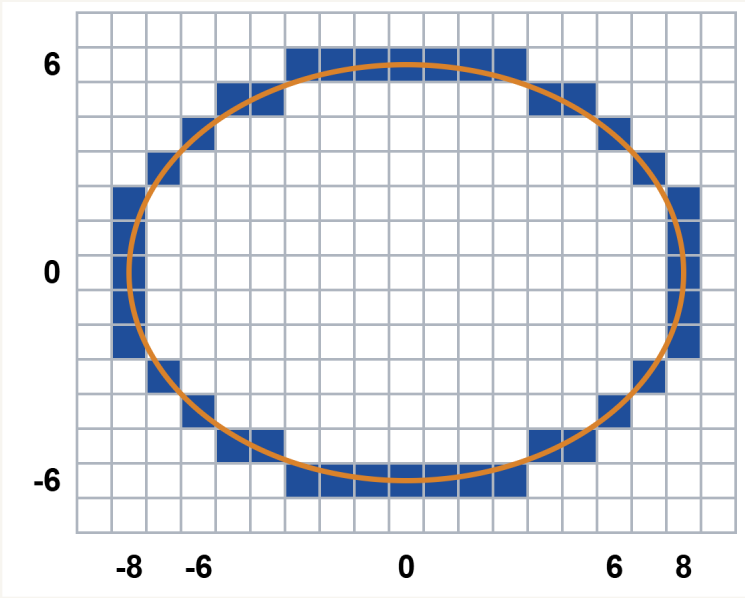
Using symmetry: the whole ellipse

Each quadrant pixel gives 4 pixels. From (4, 5):

(4, 5) (−4, 5) (4, −5) (−4, −5)

For centre (xc, yc) = (20, 15): (4, 5) → (24, 20).

11 quadrant pixels gave the whole ellipse: 40 different pixels.

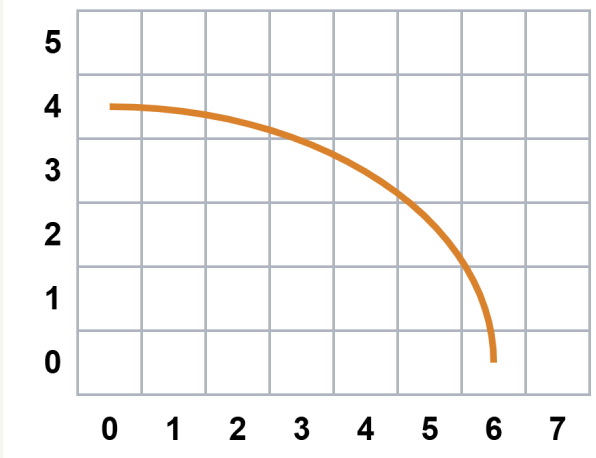


Class exercise: your turn

Use the midpoint ellipse algorithm for a = 6, b = 4, centre (0, 0).

1. Find $p1_0$, $2b^2$ and $2a^2$
2. Region 1 table until $2b^2x \geq 2a^2y$
3. Find $p2_0$ and the region 2 table
4. Shade the quadrant pixels

Time: 8 minutes.



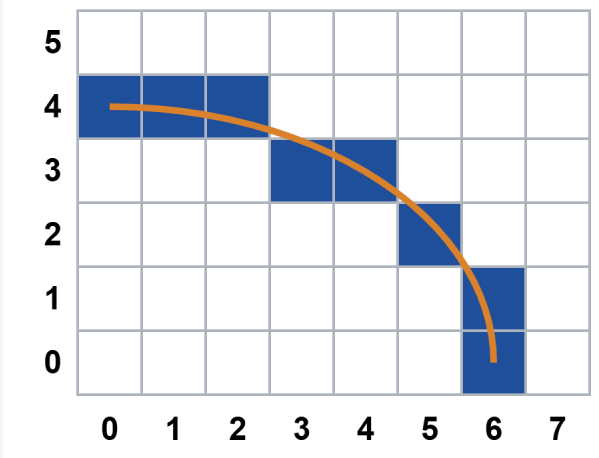
Exercise answer: a = 6, b = 4

$p1_0 = 16 - 36(4) + 9 = -119$. $2b^2 = 32$, $2a^2 = 72$.

k	$p1_k$	Decision	(x_{k+1}, y_{k+1})	$2b^2x_{k+1}$	$2a^2y_{k+1}$
0	-119	$p1 < 0 \rightarrow$ E	(1, 4)	32	288
1	-71	$p1 < 0 \rightarrow$ E	(2, 4)	64	288
2	9	$p1 \geq 0 \rightarrow$ SE	(3, 3)	96	216
3	-95	$p1 < 0 \rightarrow$ E	(4, 3)	128	216
4	49	$p1 \geq 0 \rightarrow$ SE	(5, 2)	160	144

$p2_0 = 16(5.5)^2 + 36(1)^2 - 576 = -56$

k	$p2_k$	Decision	(x_{k+1}, y_{k+1})	$2b^2x_{k+1}$	$2a^2y_{k+1}$
0	-56	$p2 \leq 0 \rightarrow$ SE	(6, 1)	192	72
1	100	$p2 > 0 \rightarrow$ S	(6, 0)	192	0



Midpoint ellipse in C++ with OpenGL

```
void midpointEllipse(int xc, int yc, int a, int b) {
    long a2 = a * a, b2 = b * b;
    int x = 0, y = b;  long px = 0, py = 2 * a2 * y;
    long p = lround(b2 - a2 * b + 0.25 * a2);            // p1
    plot4(xc, yc, x, y);
    while (px < py) {                                        // region 1
        x++;  px += 2 * b2;
        if (p < 0) p += b2 + px;
        else { y--;  py -= 2 * a2;  p += b2 + px - py; }
        plot4(xc, yc, x, y);
    }
    p = lround(b2 * (x + 0.5) * (x + 0.5) + a2 * (y - 1) * (y - 1) - a2 * b2);
    while (y > 0) {                                          // region 2
        y--;  py -= 2 * a2;
        if (p > 0) p += a2 - py;
        else { x++;  px += 2 * b2;  p += a2 - py + px; }
        plot4(xc, yc, x, y);
    }
}
```

Key points from today

1. $F(x, y) = b^2x^2 + a^2y^2 - a^2b^2$: negative inside, zero on, positive outside.
2. An ellipse has 4-way symmetry, so we compute one quadrant in two regions.
3. Region 1 (step x) while $2b^2x < 2a^2y$; region 2 (step y) until $y = 0$.
4. $p1_0 = b^2 - a^2b + \frac{1}{4}a^2$; $p2_0 = b^2(x + \frac{1}{2})^2 + a^2(y - 1)^2 - a^2b^2$.
5. Keep $2b^2x$ and $2a^2y$ as running totals: each step needs only additions.
6. Draw one quadrant at the origin, plot 4 points, then shift by (xc, yc).

Practice questions

1. Use the midpoint ellipse algorithm for a = 10, b = 6. Show both region tables.
2. Use the midpoint ellipse algorithm for a = 4, b = 2, centre (10, 10).
3. Derive $p1_0 = b^2 - a^2b + \frac{1}{4}a^2$.
4. Why does an ellipse need two regions but a circle does not?
5. Use the midpoint ellipse algorithm for a = 6, b = 8 (taller than wide). Which region is longer now?
6. Write the 4 symmetric points of (3, 5) for an ellipse centred at (20, 15).

Coming up

Next lecture

Filling a shape

Boundary fill, flood fill, 4- and 8-connected pixels, then scan-line polygon fill

Next lab

Mini Paint

Draw lines, circles and ellipses with the mouse using your own algorithms; you write the ellipse

Questions?

Thank you

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